

SUBMARINES

Submarines came into their own during World War I, and Germany, in particular, used them to excellent effect, especially against enemy merchant ships carrying vital supplies to Europe. During World War II, when Germany's surface fleet was severely limited by Hitler's prejudice against it, it was the "U-boats" that provided the Kriegsmarine with the bulk of its victories: it is a widely held view that they came close to bringing Britain to its knees. As a consequence, both Britain's Royal Navy and the US Navy were forced to make considerable investments in measures to combat the "submarine menace", which took some time to take effect. The Battle of the Atlantic could be fully declared won only after these measures had yielded results.

► RN E-CLASS

Commissioned 1913–17 **Origin** UK

Displacement 670 tons
(810 tons submerged)

Length 55.2m (181ft)

Top speed 15 knots (9 knots submerged)

Built in two groups, the British Royal Navy's *E*-class submarines were ocean-going boats with a range of 5,556km (3,000 nautical miles). At the outbreak of the war, 16 were in service or nearing completion, and 40 more were hastily constructed in no less than eight different yards. Six were constructed as minelayers; each of these were fitted with three torpedo tubes and carried 20 mines.



Flooding vents

12cm gun

Conning tower
with anti-aircraft
gun platform

Horizontal
control planes

Flood tanks

▲ U-25

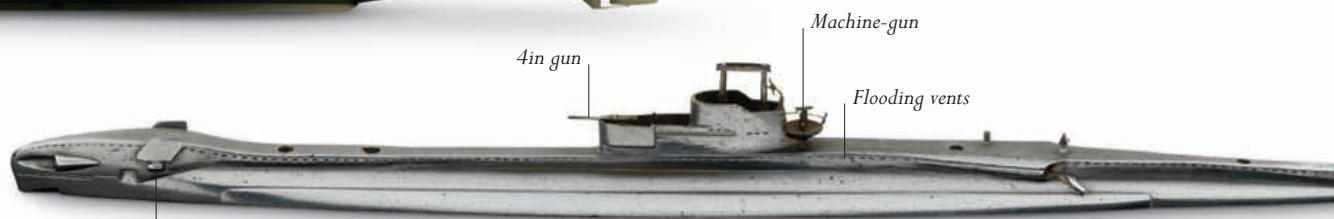
Commissioned 1936 **Origin** Germany

Displacement 860 tons
(985 tons submerged)

Length 72.4m (237½ft)

Top speed 18.5 knots
(8.3 knots submerged)

One of a pair of Type IA double-hulled ocean-going boats, the *U-25* was built by Deschimag for the Kriegsmarine in the mid-1930s, after a protracted and secretive design process. It sunk north of Terschelling, in August 1940, after hitting a mine that it had laid itself.



4in gun

Machine-gun

Flooding vents

Diving
planes

▲ RNT-CLASS (1937–44)

Commissioned 1938–45 **Origin** UK

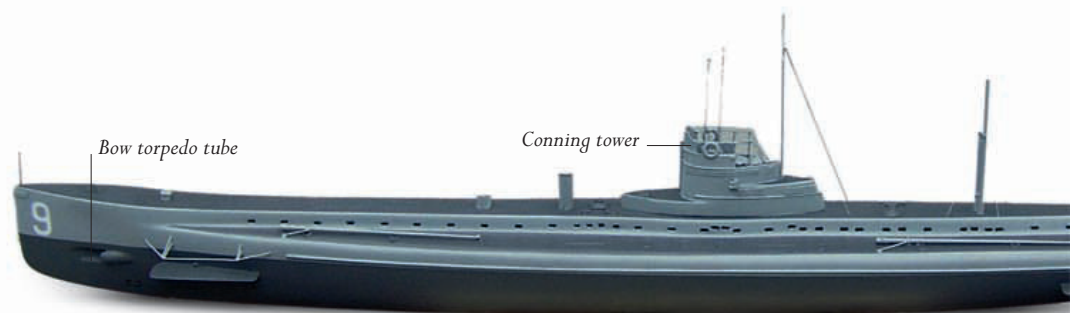
Displacement 1,320 tons
(1,575 tons submerged)

Length 83.8m (275ft)

Top speed 15.5 knots (9 knots submerged)

T-class boats were built for Britain's Royal Navy in three large groups from 1937 to 1944, undergoing considerable improvement in the process. Later builds remained in service until the 1960s. Maximum dive depth was 91m (300ft), which increased to 106m (350ft) in the last group.

Torpedo tubes



Bow torpedo tube

Conning tower

▲ U-9

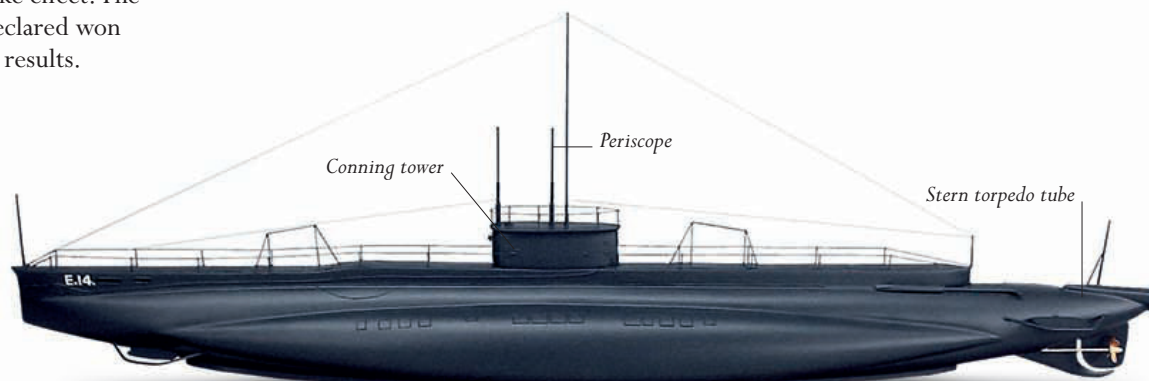
Commissioned 1910 **Origin** Germany

Displacement 495 tons
(610 tons submerged)

Length 57.4m (188½ft)

Top speed 14.2 knots (8 knots submerged)

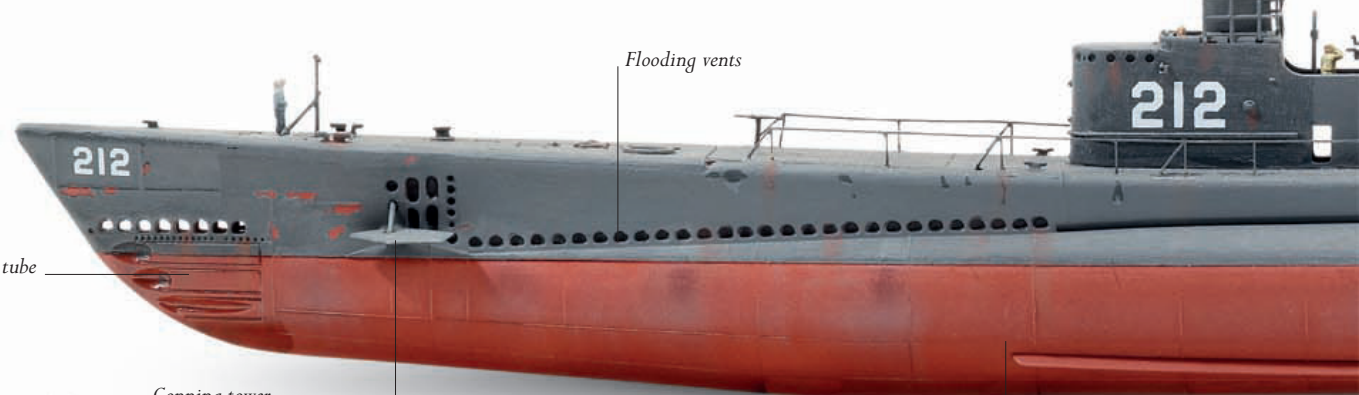
The *U-9* was leader of a class of four Type U-9 coastal submarines constructed at the Danzig Navy Yard in the Baltic. Like all German submarines built prior to 1912, they had both electric motors and petrol engines. Though these were experimental boats, they also saw combat, and all but *U-9* were war losses.



Conning tower

Periscope

Stern torpedo tube



Flooding vents

Bow torpedo tube